Autoimmune hepatitis

Speaker notes

Shubhayu Ghosh · Shyam K. Saha

6 November 2026

Notes correspond to the numbered slides in the revised presentation. Main and supplementary sections are identified separately. PDF bookmarks provide direct navigation to each slide.

Open the online presentation

Slide index

01 · Autoimmune hepatitis	2
02 · Definition	3
03 · Epidemiology	4
04 · Serological types	5
05 · Pathogenesis	6
06 · Presentation	7
07 · Clinical assessment	8
08 · Laboratory pattern	9
09 · Diagnostic approach	10
10 · Histology	11
11 · Diagnostic score	12
12 · Differential diagnosis	13
13 · Treatment	14
14 · Follow-up	15
15 · Treatment reassessment	16
16 · Special situations	17
17 · Prognosis	18
18 · Take-home	19
19 · References	20
20 · Backup A · Adult regimens	21
21 · Backup B · Variant syndromes	22
22 · Backup C · Treatment evidence	23

19 slides in the main sequence (including references); 3 supplementary slides. A reference list follows the notes.

Main presentation · Slide 01

Autoimmune hepatitis

This presentation reviews autoimmune hepatitis (AIH), including its immunopathogenesis, clinical manifestations, diagnostic evaluation and treatment. The principal clinical objective is recognition of a potentially progressive but treatable cause of hepatocellular injury.

The core sequence is intended for an undergraduate medical seminar. Supplementary slides contain adult drug regimens, variant syndromes and historical treatment evidence. N displays notes; B opens supplementary slides and returns to the main sequence.

Open slide 1 online

Main presentation · Slide 02

Definition

AIH is a chronic inflammatory liver disease of uncertain aetiology, characterised by immune-mediated hepatocyte injury, hypergammaglobulinaemia, circulating autoantibodies and compatible liver histology. Acute presentation, including acute liver failure, is recognised.

The characteristic features constitute a diagnostic constellation: none is independently diagnostic and individual features may be absent. The diagnosis requires exclusion of competing aetiologies.

Natural history: Persistent necroinflammatory activity may lead to fibrosis, cirrhosis, hepatic decompensation and death. Immunosuppressive treatment reduces inflammatory activity and improves outcomes.

Terminology: “Lupoid hepatitis” is a historical term associated with ANA positivity. AIH is a distinct disease entity rather than a manifestation of systemic lupus erythematosus.

Open slide 2 online

Main presentation · Slide 03

Epidemiology

AIH affects both sexes and all age groups, with an overall female predominance of approximately 3:1. A classical bimodal distribution in the second and fifth decades is commonly described; contemporary population studies also demonstrate substantial disease in older adults.

Frequency: EASL 2025 cites a global pooled incidence of 1.28 cases per 100,000 person-years and prevalence of 15.65 per 100,000 population. These pooled estimates vary with geography, ascertainment and study period.

Associated autoimmunity: Autoimmune thyroid disease, coeliac disease and type 1 diabetes are important associations. A personal and family history of autoimmune disease is relevant to assessment. Demographic features alone cannot exclude AIH.

Open slide 3 online

Main presentation · Slide 04

Serological types

Type 1: ANA and/or SMA are characteristic; F-actin is an important SMA target. Type 1 is the predominant form in adults, but also occurs in children. The presentation may be insidious or acute.

Type 2: Anti-LKM-1 targets cytochrome P450 2D6 (CYP2D6); anti-LC-1 targets formiminotransferase cyclodeaminase (FTCD). Type 2 predominantly affects children and adolescents and is associated with IgA deficiency and other autoimmune disease. Acute presentations are recognised.

Sex distribution: The AASLD 2024 teaching table reports approximate female:male ratios of 4:1 and 10:1. These are source-specific teaching estimates; proportions differ in paediatric cohorts and other populations.

Anti-SLA/LP: This highly specific antibody targets SepSecS and may occur with either serological profile. Historical classification of anti-SLA-positive disease as “type 3” is not retained as a separate established type.

Current classification: EASL 2025 does not recommend autoantibody-based subclassification of adults for management. Disease activity, fibrosis and treatment response are more relevant than subtype alone for clinical decisions.

Open slide 4 online

Main presentation · Slide 05

Pathogenesis

Susceptibility: HLA class II associations implicate antigen presentation in disease susceptibility. HLA-DRB1*03:01 and *04:01 are established type 1 risk alleles in European populations, but are not required for AIH development. AIRE defects in APECED illustrate defective central tolerance in a specific, uncommon clinical setting.

Initiation: Molecular mimicry and altered gut–liver signalling are proposed mechanisms. Sequence similarities involving CYP2D6 and viral proteins do not establish a particular virus as the cause of idiopathic AIH. Drug-induced autoimmune-like hepatitis requires separate clinical consideration.

Antigen presentation and regulation: Liver antigen presentation activates autoreactive T cells in a susceptible host. Regulatory T-cell frequency and function have shown heterogeneous abnormalities across studies; a universal numerical deficiency is not established.

Effector mechanisms: Th1-associated IFN-γ and Th17-associated IL-17 contribute to inflammatory recruitment and activation. Cytotoxic T-cell mechanisms include perforin/granzyme and Fas–FasL pathways. Plasma-cell differentiation contributes to IgG elevation; ANA and SMA are principally diagnostic markers.

Fibrogenesis: Persistent injury and inflammatory signalling activate hepatic stellate cells and extracellular-matrix deposition. The sequence is a conceptual framework, with components overlapping rather than occurring as a proven linear cascade in every patient.

Open slide 5 online

Main presentation · Slide 06

Presentation

AIH may be detected through incidental aminotransferase elevation or present with fatigue, malaise, arthralgia, anorexia, amenorrhoea or jaundice. An acute presentation may represent new disease or an exacerbation of previously unrecognised chronic hepatitis.

Presentation frequencies: BSG 2025 reports acute hepatitis in 25–50% and associated extrahepatic autoimmune disease in 24–42%. These ranges reflect heterogeneous cohorts and are not mutually exclusive categories.

Cirrhosis: In the Danish nationwide cohort of 1,721 patients, 1,318 underwent biopsy at diagnosis; 28.3% of the biopsied group had cirrhosis. This denominator is important when interpreting the figure.

Severity: BSG defines acute severe AIH as acute jaundice with INR >1.5 without encephalopathy. Encephalopathy in an acute presentation with coagulopathy indicates AIH-related acute liver failure. Established chronic liver disease requires separate assessment for acute-on-chronic deterioration.

Asymptomatic presentation does not imply mild histological activity. Normal IgG or initially negative conventional autoantibodies can complicate acute diagnosis.

Open slide 6 online

Main presentation · Slide 07

Clinical assessment

Examination may be normal or demonstrate jaundice, hepatomegaly, splenomegaly and stigmata of chronic liver disease. Ascites, oedema, variceal bleeding or encephalopathy indicate decompensation. Associated autoimmune disease may contribute extrahepatic findings; none of these features is specific to AIH.

Injury versus function: Aminotransferases reflect hepatocellular injury. INR and albumin assess aspects of synthetic function, while bilirubin reflects excretory dysfunction. INR is especially important in acute deterioration; albumin is also influenced by nutrition, inflammation and non-hepatic losses.

Portal hypertension: Splenomegaly and thrombocytopenia may support its presence. Assessment of decompensation and mental status takes priority over awaiting complete serological results.

Open slide 7 online

Main presentation · Slide 08

Laboratory pattern

Biochemistry: ALT and AST elevations generally predominate over ALP. Marked cholestasis requires assessment for biliary obstruction, PBC, PSC or a variant syndrome rather than attribution to classical AIH alone.

IgG: Selective polyclonal IgG elevation is characteristic. In the Hartl multicentre study, 130 of 1,318 patients (10%) had normal IgG at presentation; this did not preclude a typical treatment response. Normal values are particularly relevant to the differential diagnosis of acute disease.

Serology: ANA and SMA are not disease-specific and may occur in MASLD, other liver diseases or healthy individuals. Anti-SLA/LP has high specificity but limited sensitivity. Negative routine antibodies do not exclude AIH; extended testing and biopsy may be required.

Interpretation: Adult ANA/SMA titres of ≥1:40 and ≥1:80 carry different weights in the original simplified score. Assay substrate and validated cut-offs matter, and paediatric thresholds differ.

Reference: Hartl J et al. Features and outcome of AIH patients without elevation of IgG. JHEP Reports 2020. https://pmc.ncbi.nlm.nih.gov/articles/PMC7139106/

Open slide 8 online

Main presentation · Slide 09

Diagnostic approach

Investigation comprises clinical assessment, liver biochemistry, serum IgG, relevant autoantibodies, imaging and exclusion of alternative aetiologies. Ultrasound assesses liver morphology and biliary obstruction. A complete medication, herbal-product, alcohol and exposure history is essential.

HBV and HCV testing is part of the routine aetiological screen. Acute jaundice also warrants appropriate HAV and HEV testing; additional viral investigations depend on context. Copper studies, transferrin saturation and alpha-1 antitrypsin evaluation are selected according to age and presentation.

Liver biopsy establishes the inflammatory pattern, grades activity and stages fibrosis, while also assessing competing diagnoses. A transjugular route may be considered when percutaneous biopsy is unsuitable.

The IAIHG score supports clinical–pathological integration. Its limitations are important in acute, seronegative and variant disease. Investigation must proceed alongside urgent management when hepatic failure is suspected.

Open slide 9 online

Main presentation · Slide 10

Histology

Interface hepatitis: Inflammation extends from the portal tract across the limiting plate into adjacent hepatocytes. The infiltrate is lymphocytic or lymphoplasmacytic; plasma-cell enrichment supports AIH, but its absence does not exclude the disease. “Piecemeal necrosis” is the older term.

Acute patterns: Lobular hepatitis and centrilobular necrosis may be prominent, sometimes without the full portal-based chronic pattern. Histological interpretation must account for the clinical presentation and competing diagnoses.

Classical additional features: Hepatocyte rosettes are groups of regenerating hepatocytes arranged around a small lumen. Emperipolesis describes an intact inflammatory cell within a hepatocyte. Both lack sufficient specificity to establish AIH.

Activity and stage: The report should assess necroinflammatory activity and fibrosis. The 2022 International AIH Pathology Group consensus emphasises inflammatory pattern and alternative aetiologies. Its criteria differ from the original 2008 simplified score, which incorporated rosettes and emperipolesis in its definition of “typical” histology.

Open slide 10 online

Main presentation · Slide 11

Diagnostic score

The original simplified adult IAIHG score has four domains: autoantibodies, serum IgG, histology and exclusion of viral hepatitis. A total of ≥6 supports probable AIH and ≥7 supports definite AIH by these criteria.

Case calculation: ANA 1:160 = 2 points; IgG 1.5 × ULN = 2; typical histology = 2; viral hepatitis excluded = 2. Total: 8. Autoantibody points are capped at 2, regardless of the number of positive antibodies.

Original histological definition: Typical histology requires interface hepatitis with lymphocytic/lymphoplasmacytic portal inflammation extending into the lobule, rosettes and emperipolesis. Compatible histology lacks the complete typical constellation.

Limitations: This is the original 2008 adult score. Acute and seronegative presentations can score below threshold despite AIH, and this calculator is not intended for paediatric or overlap diagnosis. EASL 2025 discusses assay-specific updates and newer histological criteria.

Open slide 11 online

Main presentation · Slide 12

Differential diagnosis

Viral hepatitis: HBV and HCV testing is routine. Select serology and nucleic-acid assays according to presentation, timing and exposure. HAV and HEV should be evaluated in acute jaundice; positive autoantibodies do not by themselves exclude viral hepatitis.

Drug-induced autoimmune-like hepatitis: Relevant exposures include nitrofurantoin, minocycline and some herbal products. Biochemistry and histology may closely resemble idiopathic AIH. Sustained resolution after drug withdrawal and absence of relapse off immunosuppression favour drug-induced disease, but are not absolute discriminators.

Wilson disease: Evaluate younger patients using the combined clinical and copper assessment. Caeruloplasmin alone cannot exclude Wilson disease. Urinary copper, slit-lamp examination and further specialist investigations may be required.

MASLD and other aetiologies: Steatosis and low-titre ANA may be incidental or coexist with AIH. Alcohol-related disease, haemochromatosis and alpha-1 antitrypsin deficiency also warrant consideration when appropriate.

Cholestatic variants: AMA and bile-duct lesions suggest PBC; MRCP evaluates PSC when indicated. Atypical cholestasis or poor immunosuppressive response should prompt reassessment.

Open slide 12 online

Main presentation · Slide 13

Treatment

Indication: Active AIH generally requires immunosuppression, including asymptomatic disease. Symptoms are an unreliable measure of necroinflammatory activity. Selected patients with very mild disease require individual assessment of treatment benefit and toxicity.

Induction: Prednisolone produces a relatively rapid anti-inflammatory effect. EASL 2025 recommends at least 0.5 mg/kg/day, potentially up to 1 mg/kg/day in more severe disease. The dose is reduced according to biochemical response.

Steroid-sparing treatment: Azathioprine is used with corticosteroids and for maintenance, not as sole induction therapy. EASL introduces it preferably about two weeks after steroids, when bilirubin is <6 mg/dL, starting at 50 mg/day and titrating to 1–2 mg/kg/day as tolerated.

Current alternatives: EASL 2025 supports prednisolone with MMF (1.5–2 g/day) as an adult first-line option. This differs from guidance positioning MMF primarily after azathioprine intolerance. MMF is teratogenic. EASL 2025 does not recommend budesonide as first-line treatment; cirrhosis is a contraindication.

Pretreatment assessment: CBC, liver biochemistry and TPMT assessment precede azathioprine; infection risks, HBV status, vaccination, glucose/BP and bone health require review. Normal TPMT activity does not eliminate the need for CBC monitoring.

Open slide 13 online

Main presentation · Slide 14

Follow-up

Response definitions: The IAIHG 2022 consensus defines non-response as <50% reduction in aminotransferases within four weeks, and complete biochemical response as normal aminotransferases and IgG within six months. Insufficient response is failure to achieve the latter. EASL 2025 allows assessment of complete biochemical response over 6–12 months; the slide retains the original IAIHG time points.

Histological remission: The consensus definition is a modified hepatitis activity index <4/18. Biochemical normalisation does not necessarily establish histological remission, particularly in cirrhosis.

Monitoring: Review CBC and liver biochemistry frequently during initiation and dose changes, with IgG as a response marker. Stable maintenance generally permits review every 3–6 months. Monitor corticosteroid toxicity, infection and bone health.

Withdrawal: EASL 2025 permits a trial only in carefully selected patients with stable biochemical response maintained for at least two years on low-dose monotherapy. BSG 2025 suggests considering steroid-sparing drug withdrawal after 3–4 years of sustained response. Corticosteroid tapering and withdrawal of all immunosuppression are distinct decisions.

Relapse: Recurrence after withdrawal commonly requires re-induction followed by prolonged maintenance. Persistent disease activity, cirrhosis and repeated relapse adversely affect prognosis. Follow-up remains necessary after treatment cessation.

Open slide 14 online

Main presentation · Slide 15

Treatment reassessment

Inadequate response: Apply the response definitions on the preceding slide, then reassess the diagnosis, adherence, drug dose and concurrent liver disease. Review whether corticosteroids were tapered too rapidly. Thiopurine metabolites, when available, can help distinguish underexposure, non-adherence and toxicity.

Failure to reach a biochemical target does not automatically require an immediate change of drug. Persistent aminotransferase elevation, cirrhosis or clinical deterioration strengthen the need for specialist escalation. Acute severe AIH requires reassessment over days rather than the chronic-treatment time points.

Intolerance: An adverse effect that prevents continued treatment is distinct from failure to suppress the disease. MMF is an established alternative for azathioprine intolerance, with counselling about teratogenicity. True insufficient responders may need optimisation or other specialist-directed therapy, including tacrolimus in selected cases.

Relapse: Recurrent inflammatory activity after treatment withdrawal should prompt exclusion of competing causes of liver injury. Re-induction with corticosteroid-based treatment is usually followed by prolonged maintenance. Activity recurring during treatment tapering or maintenance is often termed a flare; adherence and treatment exposure should be reviewed.

Open slide 15 online

Main presentation · Slide 16

Special situations

Acute severe AIH: BSG defines acute jaundice with INR >1.5 and no encephalopathy. A corticosteroid trial may be appropriate after assessment for infection, in close liaison with a transplant centre. Clinical deterioration requires immediate reassessment; absence of improvement over 3–7 days is an escalation signal.

Acute liver failure: Encephalopathy accompanying acute coagulopathic hepatitis requires urgent transplant-centre assessment. Referral should not await completion of a predetermined steroid course. Acute-on-chronic liver failure is a separate presentation in established chronic liver disease.

Pregnancy: Disease control should be optimised before conception. Prednisolone and azathioprine may be continued when indicated. MMF is contraindicated in pregnancy and requires planned preconception substitution; EASL advises withdrawal at least three months before conception. Postpartum flares warrant close follow-up.

Cirrhosis: Assess portal hypertension, decompensation and transplant indications. HCC surveillance uses six-monthly ultrasound, with or without AFP according to protocol. Budesonide is contraindicated. Biochemical response does not eliminate cirrhosis-related surveillance requirements.

Open slide 16 online

Main presentation · Slide 17

Prognosis

Disease stage: Cirrhosis at diagnosis or its subsequent development is an adverse prognostic marker. Decompensation and portal hypertension identify patients with greater risk of liver-related events and may require transplant assessment. Compensated cirrhosis can still respond to immunosuppression.

Treatment response: Complete biochemical response within six months, particularly when sustained, is associated with lower subsequent liver-related death, transplantation and other liver events. These are prognostic associations rather than guarantees for an individual patient. Biochemical response does not establish histological remission.

Cumulative activity: Delayed biochemical control, persistent inflammation and repeated relapses are associated with progressive fibrosis and worse long-term outcomes. Adherence and treatment tolerance therefore affect the ability to maintain control.

Follow-up: Remission reduces risk but does not eliminate the consequences of established cirrhosis. Continue appropriate portal-hypertension assessment and six-monthly HCC surveillance. Monitor for relapse even after supervised treatment withdrawal. Historical untreated-disease mortality figures should not be used as a contemporary prognosis.

Open slide 17 online

Main presentation · Slide 18

Take-home

AIH should remain in the differential diagnosis of otherwise unexplained hepatocellular injury and cryptogenic cirrhosis. Diagnosis is based on the combined clinical, biochemical, serological and histological assessment.

Treatment aims to suppress active inflammation, achieve biochemical response and prevent progressive fibrosis. Relapse, drug toxicity and the complications of established cirrhosis determine the requirements for long-term care.

Open slide 18 online

Main presentation · Slide 19

References

The principal sources are the EASL 2025 and BSG 2025 clinical practice guidelines, the AASLD 2019 practice guidance, and the original IAIHG diagnostic and response criteria. The AASLD 2024 teaching resource supplies the explicitly attributed subtype sex ratios.

The 2022 histological consensus should be distinguished from the original 2008 scoring definition. EASL and BSG differ in some drug choices, response time frames and withdrawal criteria; these differences are identified in the relevant notes.

Additional primary evidence: Hartl et al., Features and outcome of AIH patients without elevation of IgG, JHEP Reports 2020, https://pmc.ncbi.nlm.nih.gov/articles/PMC7139106/; Grønbæk et al., J Hepatol 2014;60:612–617, doi:10.1016/j.jhep.2013.10.020.

Supplementary slides provide adult regimens, variant-syndrome assessment and historical treatment-trial results.

Open slide 19 online

Supplementary · Slide 20

Backup A · Adult regimens

Prednisolone: EASL 2025 recommends at least 0.5 mg/kg/day and potentially up to 1 mg/kg/day in more severe disease, followed by a response-guided taper. BSG 2025 generally favours approximately 0.5 mg/kg/day, usually not exceeding 40 mg/day.

Azathioprine: EASL recommends introduction preferably about two weeks after corticosteroids when bilirubin is <6 mg/dL, starting at 50 mg/day and increasing to 1–2 mg/kg/day. BSG starts at approximately 1 mg/kg/day. CBC monitoring remains essential despite normal TPMT activity.

MMF: EASL 2025 supports an adult first-line combination with prednisolone, with a target dose of 1.5–2 g/day. MMF is also used for azathioprine intolerance. Teratogenicity requires counselling, effective contraception and preconception planning.

Toxicity: Corticosteroids cause metabolic, skeletal, ocular and infective complications. Azathioprine may cause myelosuppression, hepatotoxicity, gastrointestinal intolerance or pancreatitis. Significant toxicity warrants prompt review and treatment adjustment.

Further therapy: Tacrolimus is a specialist option for selected insufficient responders. Budesonide is contraindicated in cirrhosis and is not a first-line recommendation in EASL 2025. These regimens apply to adults; paediatric dosing and monitoring differ.

Open slide 20 online

Supplementary · Slide 21

Backup B · Variant syndromes

Variant syndromes combine hepatitic and cholestatic features. PBC denotes primary biliary cholangitis; PSC denotes primary sclerosing cholangitis; UDCA denotes ursodeoxycholic acid. Neither an isolated positive AMA nor a simplified AIH score independently establishes overlap.

AIH–PBC: Assess cholestatic biochemistry, AMA and bile-duct injury alongside the severity of interface hepatitis. The predominant component and inflammatory activity guide treatment; UDCA with immunosuppression is appropriate in selected patients.

Paris criteria: Classically ≥2 PBC features (ALP ≥2 × ULN or GGT ≥5 × ULN; AMA; florid duct lesion) and ≥2 AIH features (ALT ≥5 × ULN; IgG ≥2 × ULN or SMA; moderate/severe interface hepatitis), with interface hepatitis required. Some variants do not fulfil these criteria.

AIH–PSC: MRCP is indicated when cholestasis, pruritus, IBD, biliary biopsy changes or inadequate response suggest duct disease. Sclerosing cholangitis requires particular consideration in childhood AIH. Immunosuppression targets the hepatitic component rather than reliably reversing established PSC duct disease.

Open slide 21 online

Supplementary · Slide 22

Backup C · Treatment evidence

These historical trials established a survival benefit from corticosteroid-based treatment in severe chronic active liver disease. Their mortality estimates must be interpreted in the context of the selected populations and treatment era.

The Mayo trial demonstrated lower mortality with prednisone, alone or combined with azathioprine, than with placebo or azathioprine alone. The delayed action of azathioprine explains its role in combination and maintenance rather than sole induction therapy.

The Royal Free trial demonstrated improved long-term survival with prednisolone. Contemporary prognosis depends on sustained disease control, fibrosis stage, cirrhosis and comorbidity.

References: Soloway et al., Gastroenterology 1972;63:820–833; Kirk et al., Gut 1980;21:78–83. The cirrhosis proportion is from the Danish cohort: Grønbæk et al., J Hepatol 2014;60:612–617.

Open slide 22 online

Reference links

AASLD Liver Fellow Network. Back to Basics: ANA-lyzing Autoimmune Hepatitis. 2024. Teaching comparison and diagnostic framework.
https://www.aasld.org/liver-fellow-network/core-series/back-basics/back-basics-ana-lyzing-autoimmune-hepatitis

EASL Clinical Practice Guidelines. J Hepatol 2025;83:453–501.
https://easlcampus.eu/sites/default/files/2025-08/EASL_CPGs_on_the_management_of_AIH.pdf

Gleeson D et al. BSG guidelines. Gut 2025;74:1364–1409.
https://pmc.ncbi.nlm.nih.gov/articles/PMC12421125/

Mack CL et al. AASLD practice guidance 2019. Hepatology 2020;72:671–722.
https://doi.org/10.1002/hep.31065

Hennes EM et al. Simplified diagnostic criteria. Hepatology 2008;48:169–176.
https://doi.org/10.1002/hep.22322

Lohse AW et al. Histological consensus. Liver Int 2022;42:1058–1069.
https://doi.org/10.1111/liv.15217

Pape S et al. IAIHG response criteria. J Hepatol 2022;76:841–849.
https://doi.org/10.1016/j.jhep.2021.12.041

Mieli-Vergani G et al. Autoimmune hepatitis. Nat Rev Dis Primers 2018;4:18017.
https://doi.org/10.1038/nrdp.2018.17

Hartl et al. Features and outcome of AIH patients without elevation of IgG. JHEP Reports, 2020.
https://pmc.ncbi.nlm.nih.gov/articles/PMC7139106/

Grønbæk et al. Autoimmune hepatitis in Denmark. Journal of Hepatology 2014;60:612–617.
https://pubmed.ncbi.nlm.nih.gov/24326217/